

* Performance Metrics, Accuracy, Precision, Recall and F-Beta.

Dataset

I/P		O/P	
x_1	x_2	y	$\hat{y}$
-	-	0	1
-	-	1	1
-	-	0	0
-	-	1	1
-	-	0	1
-	-	1	0

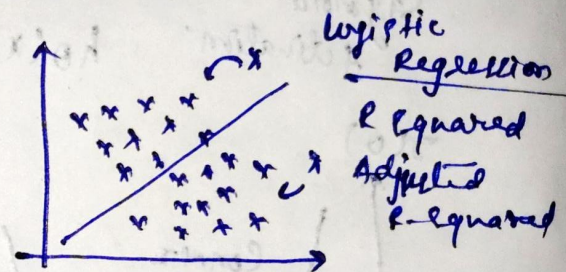
→ Predicted by model.

1. Confusion Matrix

Actual value

	0	1
1	3	2
0	1	1

Predicted value

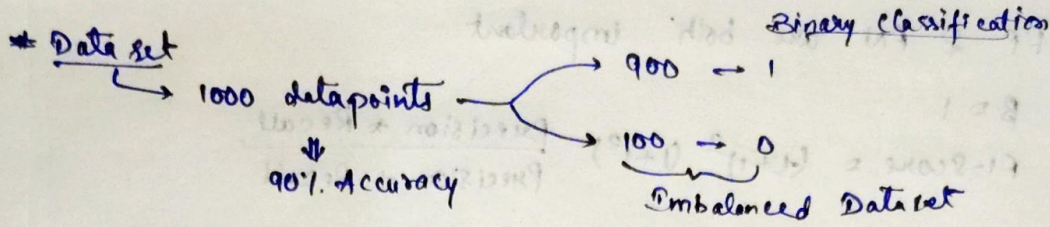


Actual

	1	0
1	TP	FP
0	FN	TN

Predicted

ii. Accuracy: $\frac{TP + TN}{TP + FP + TN + FN} = \frac{3 + 1}{3 + 2 + 1 + 1} = \frac{4}{7}$



iii. Precision: $\frac{TP}{TP + FP}$ ⇒ Out of all the actual value how many are correctly predicted

iv. Recall: $\frac{TP}{TP + FN}$ ⇒ Out of all the predicted value how many are correctly predicted

	Actual	
Predicted	0	1
	TP	FP
0	FN	TN

* Use Cases:

i. Spam classification

	Actual	
Predicted	Spam	Not Spam
	TP	FP
Not Spam	FN	TN

$\frac{TP}{TP + FP}$

Mail → Spam } Good

Model → Spam

Mail → Not Spam } Blunder

Model → Spam

ii. To predict whether a person has diabetes or not

Truth → diabetes } Blunder

Model → No diabetes

Truth → diabetes } Good

Model → diabetes

Truth → No diabetes } 2nd option

Model → diabetes } check

	Actual	
Predicted	Diabetes	No Diabetes
	TP	FP
No diabetes	FN	TN

use cases of diseases

$\text{Recall} = \frac{TP}{TP + FN}$

iii. Tomorrow the stock market will crash or not

Reducing FP ↓ or FN ↓

We use recall in this case

Reduce: FN ↓

	Actual	
Predicted	crash	Not crash
	TP	FP
Not crash	FN	TN

v. f- Beta Score: $(1 + \beta^2) \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$

① If FP & FN are both important

$$\beta = 1$$

$$F1\text{-Score} = (1+1^2) \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F1\text{-Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \rightarrow \text{Harmonic Mean}$$

② If FP is more important than FN.

$$\beta = 0.5$$

$$F0.5\text{-Score} = (1 + 0.25) \frac{P * R}{P + R}$$

③ If FN is more important than FP.

$$\beta = 2$$

$$F2\text{ Score} = (1 + 4) \frac{P * R}{P + R}$$

TP	FP
FN	TN